Fizică - curs 5

$$\begin{cases} x' = \dfrac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}} \\ y' = y \\ z' = z \\ t' = \dfrac{t - \frac{v}{c^2}x}{\sqrt{1 - \frac{v^2}{c^2}}} \end{cases}$$

$$u_x = \frac{dx}{dt}$$

$$u_x' = \frac{dx'}{dt'} = \frac{u_x - v}{1 - \frac{v u_x}{c^2}}$$

$$u_y' = \frac{dy'}{dt'} = \frac{dy}{dt - \frac{v}{c^2}dx} = \frac{u_y\sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c^2}u_x}$$

$$u_z' = \frac{u_x\sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c^2}u_x}$$

(x_1, x_2, x_3, x_4) cuadrivector de poziție

(x, y, z, ct)

$$u_x = \frac{dx}{dt} = \frac{dx}{dt}\left(\sqrt{1 - \frac{u^2}{c^2}}\right)^{-1} = \gamma_u u_x$$

dt → timp propriu cel mai mic

$$dt_0 = dt\sqrt{1 - \frac{u^2}{c^2}}$$

$$u_i = \frac{dx_i}{d\tau} = \begin{cases} u_1 = \gamma_u u_x \\ u_2 = \dfrac{dx_2}{d\tau} = \gamma_u u_y \\ u_3 = \ldots = \gamma_u u_z \\ u_4 = \gamma_u c \end{cases}$$

$$\boxed{ds^2 = (dx^2) + (dy)^2 + (dz)^2 - c^2dt^2}$$ gr. MINKOWSKI

Aplicație

Efectul Doppler relativist

$$\nu = \nu_0 \sqrt{\frac{1-\beta}{1+\beta}} \qquad \beta = \frac{v}{c}$$

↓

sursa se îndep. (V)
-//- se apropie (-V)

Dacă $v << 0 \Rightarrow \beta << 1$ — RADARE

$$\nu \cong \nu_0 (1-\beta) \Rightarrow \boxed{c \frac{|\nu - \nu_0|}{\nu_0} = v}$$

$\Delta\lambda > 0 \Rightarrow$ deplasare spre roșu $\Rightarrow$ univers în expansiune

$\Delta\lambda < 0 \Rightarrow$ —//— albastru

Noțiuni de dinamică relativistă

1. Forță în TRR
2. Energia relativistă. Relația lui EINSTEIN
3. Aplicații
4. Relația dintre impuls și energie

1. $\vec{F} = m \cdot \vec{a}$ - doar în fizica clasică

$$F = m \frac{d\vec{v}}{dt} = \frac{d(m\vec{v})}{dt} = \boxed{\frac{d\vec{p}}{dt}} \text{ ! valabilă clasic și în TRR}$$

$$\vec{p} = m\vec{v} = \frac{m_0 \vec{v}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = \frac{dm}{dt} \cdot \vec{v} + \boxed{m \frac{d\vec{v}}{dt}} \quad \vec{a}$$

$$\vec{a} = \frac{\vec{F}}{m} - \frac{1}{m}\frac{dm}{dt}\vec{v} = \frac{\vec{F}}{m} - \frac{c^2}{c^2}\cdot\frac{1}{m}\cdot\frac{dm}{dt}\vec{v}$$

$$E = mc^2$$

$$\vec{a} = \frac{\vec{F}}{m} - \frac{1}{E}\cdot\frac{dE}{dt}\vec{v}$$

$$\frac{dE}{dt} = \frac{\vec{F}\cdot d\vec{r}}{dt} = \vec{F}\cdot\vec{v} = \text{Putere}$$

$$\Rightarrow \vec{a} = \frac{\vec{F}}{m} - \frac{1}{E}(\vec{F}\cdot\vec{v})\vec{v}$$

Cazuri particulare

a) $\vec{F} \perp \vec{v}$

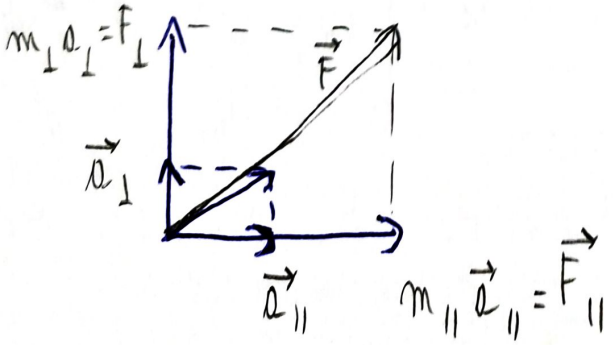
$$\vec{a}_\perp = \frac{\vec{F}_\perp}{m_\perp} = \frac{F}{\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}}$$

b) $\vec{F} \parallel \vec{v}$

$$\vec{a} = \frac{\vec{F}}{m} - \frac{\vec{F}}{E}v^2 = \frac{\vec{F}}{\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}} - \frac{\vec{F}}{\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}}\cdot\frac{v^2}{c^2} =$$

($E \to mc^2$)

$$= \frac{F_0}{m_\parallel}; \quad m_\parallel = \frac{m_0}{\left(1-\frac{v^2}{c^2}\right)^{3/2}}$$



$$dL = \vec{F}\cdot d\vec{r} = \frac{d\vec{p}}{dt}\cdot d\vec{r} = d\vec{p}\cdot\vec{v} = d(\vec{p}\cdot\vec{v}) - \vec{p}\cdot d\vec{v}$$

$$L = \int_A^B dL = \int_A^B \left[d(\vec{p}\cdot\vec{v}) - \vec{p}\,d\vec{v}\right] = \vec{p}\cdot\vec{v}\Big|_A^B - \int_A^B \frac{m_0\vec{v}\,d\vec{v}}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$d\left(\sqrt{1-\frac{v^2}{c^2}}\right) = d\left(\left(1-\frac{v^2}{c^2}\right)^{1/2}\right) = \frac{1}{2}\left(1-\frac{v^2}{c^2}\right)^{-\frac{1}{2}}\left(-\frac{2\vec{v}}{c^2}d\vec{v}\right) =$$

$$= -\frac{1}{c^2}\frac{\vec{v}\,d\vec{v}}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$L_{AB} = \vec{p}\,\vec{v}\Big|_A^B + m_0c^2\sqrt{1-\frac{v^2}{c^2}}\Big|_A^B =$$

$$= \left[\frac{m_0v^2}{\sqrt{1-\frac{v^2}{c^2}}} - m_0c^2\sqrt{1-\frac{v^2}{c^2}}\right]\Bigg|_A^B = \underbrace{m(B)c^2}_{\frac{m_0}{\sqrt{1-\frac{v(B)^2}{c^2}}}} - \underbrace{m(A)c^2}_{\frac{m_0}{\sqrt{1-\frac{v(A)^2}{c^2}}}} = \Delta m\,c^2$$

$$L_{AB} = \Delta E = \Delta m\cdot c^2$$

$$E = mc^2 = \frac{m_0c^2}{\sqrt{1-\frac{v^2}{c^2}}}$$

$v \ll c$

$$\left(1-\frac{v^2}{c^2}\right)^{-\frac{1}{2}} \cong 1 + \frac{1}{2}\frac{v^2}{c^2}$$

$$E \cong m_0\left(1+\frac{1}{2}\frac{v^2}{c^2}\right)c^2$$

$$E \cong m_0c^2 + \frac{m_0v^2}{2} \rightarrow E_{cin.}\ \text{relativistă}$$

$$E_{cin} = E - m_0 c^2 = (m - m_0) c^2$$

Aplicații

a) reacții de creare și anihilare de particule

$$e^-(\text{rapid}) + e^-(\text{repaus}) \longrightarrow e^- + e^- + e^- + e^+$$

$$\Delta E + 2m_0c^2 \longrightarrow 4m_0c^2$$

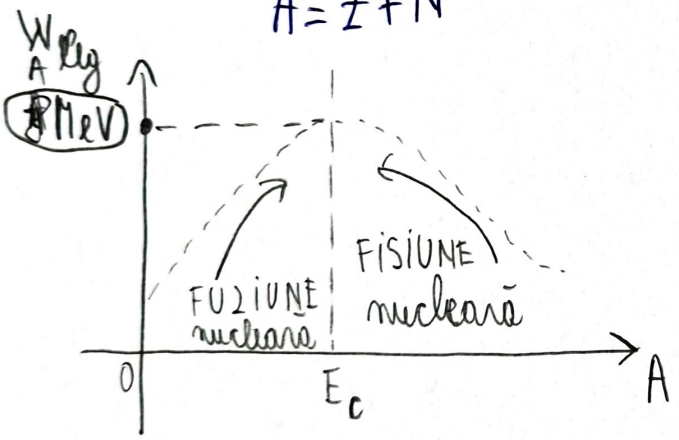
b) R de anihilare (particulă + antiparticule)

$$e^- + e^+ \longrightarrow 2\gamma \longrightarrow \text{PET}$$

c) defectul de masă

$$W_{leg} = \Delta m \cdot c^2 = [Z m_p + (A-Z) m_n - m(Z,A)] c^2$$

$$A = Z + N$$



Ex. $^2_1H + ^2_1H + ^4_2He + 23,9\ MeV$ fuziune

$^{236}_{92}U + ^{90}_{37}Rb + ^{143}_{55}Ce + 3^1_0n$ fisiune

④ $E = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2 \cdot m^2}{c^2 \cdot m^2}}} \cdot p^2$

$\frac{E^2}{c^2}$

$E = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2 m^2}{c^2 m^2}}}$

$v^2 m^2 = p^2$

$c^2 m^2 = \frac{E^2}{c^2}$

$E^2 = \frac{m_0^2 c^4}{1 - \frac{p^2}{m^2 c^2}}$

$E^2 = p^2 c^2 + m_0^2 c^4$

Ex. Det. $\frac{m_0}{m} = ?$ știind E_{cin} și p.

$E_{cin} = (m - m_0) c^2$ + E-masat

$E^2 = p^2 c^2 + m_0^2 c^4$